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Project Prototype for STAT UN3016 Applied Machine Learning

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Background and Prior Work

In the past decades, rising heat has developed into a nationwide concern in the United States. As the Environmental Protection Agency reports, global warming has contributed to average temperatures increasing across the country by about 60% since the 1970s[1]. This worsening climate has severe public health implications, with heat stress being the "leading cause of weather-related deaths" according to the World Health Organization, as it exacerbates pre-existing illnesses and increases the likelihood of infectious diseases being spread[2]. In the U.S. specifically, a recent study conducted by the Yale School of Public Health found that heat-exposure related deaths rose by over 50% in the past twenty years, with the annual average increasing from 2,670 between 2000 and 2009 to over 4,000 between 2010 and 2020 [3].

Large metropolitan areas are particularly susceptible to this trend because of the "heat island effect"[4]. Scholars describe this phenomenon as the tendency of heat to accumulate in cities because of how buildings, roads, and other manmade structures absorb warmth to a significantly greater degree than natural surfaces do[5]. Furthermore, many urban areas have much less green space per person than recommended because of the housing and corporate development

[1] The Environmental Protection Agency, "Heat Island Trends | US EPA," EPA, 2026, https://www.epa.gov/heatislands/heat-island-trends
[2] The World Health Organization, "Heat and health." WHO, 2024, https://www.who.int/news-room/fact-sheets/detail/climate-change-heat-and-health
[3] Colin Poitras, "Warming U.S. climate linked to rising deaths from heat," Yale School of Public Health, November 7th, 2025, https://ysph.yale.edu/news-article/warming-us-climate-linked-to-rising-deaths-from-heat/
[4] WHO, "Heat and health."
[5] Jeremy Gregroy and Hessam Azarijafari, "Urban Heat Islands", MIT Climate Portal, April 16, 2021, https://climate.mit.edu/explainers/urban-heat-islands

necessary to support such a large population, reducing the cooling effect from green space[6]. New York City faces this effect more acutely than anywhere else in the nation, as a study conducted by Climate Central reports[7]. Residents experience temperatures around 9.7 degrees hotter than otherwise, the largest "per capita temperature boost" in comparison to the 65 other cities analyzed in the survey[8]. Other data serves to corroborate the danger such climate inevitably poses for New Yorkers at large, with the city reporting that, on average, more than 500 residents die every summer as a result of the heat[9]. Consequently, the city devotes a considerable amount of resources towards reducing the effects of the heat as much as they are able to. These include Cooling Centers, for example, which are air-conditioned public areas that people can access for temporary respite if temperatures are higher than 95 degrees for two or more days or 100 degrees at any point in time[10].

Crucially, though, as the Climate Central report indicates, not all of the areas of the city are affected equally by the heat[11]. Indeed, some parts of the city have indexes up to 13 degrees, while others experience less than 6 degrees[12]. Similarly, not every individual experiences the severity of the heat in the same way, with research finding elderly, socially-isolated, low-income, and non-white populations in New York City to be the most at-risk to heat-related health problems[13]. Given this asymmetry, it follows that a clear methodology must be implemented to

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[6] Madeleine Gregory, "NASA Data Reveals Role of Green Spaces in Cooling Cities," NASA Science, November 26, 2024, https://science.nasa.gov/earth/climate-change/nasa-data-reveals-role-of-green-spaces-in-cooling-cities/.
[7] Climate Central, "Urban Heat Hot Spots in 65 Cities.," Climate Central, July 9, 2024, https://www.climatecentral.org/climate-matters/urban-heat-islands-2024
[8] Climate Central, "Urban Heat Hot Spots"
[9] New York City Department of Health and Mental Hygiene, "2025 NYC Heat-Related Mortality Report," *Environment and Health Data Portal*, accessed March 30, 2026, https://a816-dohbesp.nyc.gov/IndicatorPublic/data-features/heat-report/
[10] "Cooling Centers," NYC311, accessed March 30, 2026, https://portal.311.nyc.gov/article/?kanumber=KA-02663
[11] Climate Central, "Urban Heat Hot Spots"
[12] Caroline Lewis, "The 'urban heat island' effect is making New Yorkers hotter, study finds," Gothamist, July 13, 2024, https://gothamist.com/news/the-urban-heat-island-effect-is-making-new-yorkers-hotter-study-finds
[13] Joyce Klein Rosenthal et al., "Intra-urban vulnerability to heat-related mortality in New York City, 1997–2006," *Health Place* 30 (2014), 1https://pmc.ncbi.nlm.nih.gov/articles/PMC4348023/; EPA, "Heat Island Trends | US EPA"

determine where in the city citizens are most vulnerable, so that the government can allocate more resources there during periods of strong heat. This is the role of the Heat Vulnerability Index (HVI), a statistical measure which identifies the neighborhoods within a city whose residents are most at risk of fatalities due to the heat[14]. HVIs emerged in the early to mid-2000s, building upon scholarship in the intersection between societal marginalization and environmental concerns, such as Susan Cutter's Social Vulnerability Index[15]. They quickly proliferated across the nation and, indeed, around the world, with studies emerging in the United Kingdom, Korea, Canada, and China[16].

HVIs are developed using a variety of factors. Frequently, these elements can be categorized as thermal/environmental or social/demographic[17]. For example, an HVI developed for Philadelphia included factors such as surface temperature, ethnicity, income, and age, as reviewed by David Hondula et al[18]. In Phoenix, an HVI reviewed by Winston Chow et al. used similar elements, but supplemented them with immigration status and vegetation levels[19]. The geographical unit of analysis also varies between the different indices, whether it be census block groups, counties, census tracts, or postal codes[20]. New York City's HVI was created by the city's

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[14]New York City Department of Health and Mental Hygiene, "Heat Vulnerability Index," Environment and Health Data Portal, accessed March 30, 2026, https://a816-dohbesp.nyc.gov/IndicatorPublic/data-features/hvi/

[15]Susan L. Cutter, Bryan J. Boruff, and W. Lynn Shirley, "Social Vulnerability to Environmental Hazards," Social Science Quarterly 84, no. 2 (2003): 242–261, https://doi.org/10.1111/1540-6237.8402002; Yujie Niu et al., "Major Factors, Methods, and Spatial Units of Heat Vulnerability Assessment: A Systematic Review," International Journal of Environmental Research and Public Health 18, no. 20 (2021), https://pmc.ncbi.nlm.nih.gov/articles/PMC8531084/

[16]Niu et al., "Heat Vulnerability Assessment."

[17]Niu et al., "Heat Vulnerability Assessment."; Jonatan K. V. et al., "The Construction and Validation of the Heat Vulnerability Index: A Review," International Journal of Environmental Research and Public Health 12, no. 7 (2015): 7220–7234, https://doi.org/10.3390/ijerph120707220

[18] David M. Hondula et al., "Fine-Scale Spatial Variability of Heat-Related Mortality in Philadelphia County, USA, from 1983–2008: A Case-Series Analysis," Environmental Health 11, no. 1 (2012): Article 16, https://doi.org/10.1186/1476-069X-11-16

[19]Winston T. L. Chow, Wen-Ching Chuang, and Patricia Gober, "Vulnerability to Extreme Heat in Metropolitan Phoenix: Spatial, Temporal, and Demographic Dimensions," Professional Geographer 64, no. 2 (2012): 286–302, https://doi.org/10.1080/00330124.2011.600225 .

[20]Niu et al., "Heat Vulnerability Assessment."

Department of Health and Mental Hygiene, and analyzes data on surface temperature, green space, home air conditioning, and income in order to determine the score of a particular ZIP code[21]. A ranking of 1 represents the lowest vulnerability, while 5 is associated with the highest[22].

Since its inception, the NYC HVI has enabled the city to focus resources towards communities that have been deemed highly vulnerable by the index. For example, they have concentrated outreach efforts towards these neighborhoods in order to provide clarity on the health risks with heat[23]. They have also begun installing roofs lined with vegetation, along with planting more trees through the "Cool Neighborhoods NYC" initiative[24]. They also developed "Cool It! NYC," a citywide plan to increase the amount of publicly available "cooling features" in areas with high Heat Vulnerability, including pools, showers, and drinking fountains[25]. There are a few limitations to the NYC HVI, though, which researchers have endeavored to address over the years. For example, academics developed an "Enhanced Heat Vulnerability Index" which builds upon the existing framework by adding street tree data, park properties, and age data[26]. Despite these improvements, a clear lacuna remains: *the HVI is a static measure.* Scholars agree that, while useful when data is available and compiled throughout the years, the index considers vulnerability in hindsight rather than constructing a dynamic, real, and predictive picture. More detailed models, such as the "Enhanced Heat Vulnerability Index," incorporate

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[21]New York City Department of Health and Mental Hygiene, "Heat Vulnerability Index."
[22] Sophia Goertz, "Development of an Enhanced Heat Vulnerability Index for New York City to Find Priority Neighborhoods for Tree Planting," Medium, December 5, 2025, https://medium.com/@sg9437/development-of-an-enhanced-heat-vulnerability-index-for-new-york-city-to-find-priority-cee66cec741d
[23]"Extreme Heat," Hazard Profile, NYC Hazard Mitigation Plan, accessed March 30, 2026, https://nychazardmitigation.com/documentation/hazard-profiles/extreme-heat/.
[24]NYC Hazard Mitigation Plan. "Extreme Heat."
[25]NYC Department of Parks & Recreation, "Cool It NYC," Health and Safety Guide, accessed March 30, 2026, https://www.nycgovparks.org/about/health-and-safety-guide/cool-it-nyc.
[26] Goertz, "Enhanced Heat Vulnerability Index."

more granular elements, but they still fail to capture real-time fluctuations in risk. It is within this gap that our project intervenes by using machine learning trained on daily behavioral and environmental signals to transform the HVI from a fixed index into a living tool.

Research Questions and Hypotheses

Building on this gap in the existing literature, our project investigates the following research question: *can ML models trained on daily behavioral and environmental signals predict and track temporal variation in heat vulnerability across NYC neighborhoods?* Our research question stems from the need to address acute heat risks and vulnerabilities not just on yesterday's data, but on dynamic data inputs. Because heat risk is not constant, our project contributes to this literature by using machine learning techniques to transform the HVI metric into a real-time tool.

To answer this research question, we developed two primary hypotheses, which we outline below:

1. Models trained on combined daily behavioral and environmental signals (including 311 heat complaints, temperature, tree coverage, population density, population, water fountains, and energy usage) will be able to predict established HVI scores by zip code for 2024.
2. The temporal HVI scores by zip code predicted by the three ML models will be different from the HVI scores by zip code calculated by the NYCDOHMH in 2024, indicating that temporal patterns not captured by HVI are important determinants of heat vulnerability.

While these two hypotheses may seem contradictory—predicting HVI scores correctly but also predicting different HVI scores—this is not the case. This is because the mean HVI scores that the models predict are hypothesized to be the same as the ground-truth HVI scores, but the temporal HVI scores are not. This is the heart of our research: capturing the micro-scale, day-to-day changes in HVI score so as to avoid dangerous generalizations and help people get the resources they need when they need it.

Machine Learning Model

Methodology Overview

We use three different ML models to address these hypotheses. Specifically, we use decision trees, random forest models, and the XGBoost algorithm. These specific models were chosen purposefully as they build on each other.

Decision trees are the most computationally simple of the three models, and we use them to construct our preliminary output HVI scores. Specifically, decision trees work by using a hierarchical tree structure, where an internal node represents a feature, the branch represents a decision rule, and each leaf node represents the outcome of the dataset. In our case, this would look like nodes representing each of our variables (e.g., the number of 311 calls on a given day per ZIP code, the number of trees in a ZIP code, etc.), branches representing rules like "is the number of trees in an area greater than x?", and leaf nodes representing each of the HVI score categories.

Because decision trees are prone to overfitting, they might not be as accurate as we need them to be. So, we also create a random forest model. Random forests are an ensemble method that bags (bootstrap aggregating, a resampling technique) individual decision trees to improve predictive capacity. Specifically, each tree in a random forest is trained independently of all the other trees to make them strong in their own right, and then each of these trees is joined with the others to make a model that is ideally stronger than just an individual decision tree.

Finally, we use XGBoost, which uses gradient descent to create highly precise and accurate models. It works by first making predictions off of a single, poor-performing decision tree, and progressively adding trees to the forest that each learns from the prior trees' predictive mistakes. More specifically, the algorithm calculates the residuals of each tree's predictions to determine how far off the model's predictions were from reality. Residuals are the difference between the model's predicted and actual values. The residuals are then aggregated to score the model with a loss function, and in our case, the loss function is cross-entropy loss since we are doing a classification problem. XGBoost is a very popular model, in part because of its delicate handling of missing or anomalously-correlated values. This is especially useful in our project, which uses both fixed and temporal data and therefore includes many repeat values.

Creating the Dataset

The three models were created using seven datasets representative of a wide variety of socioeconomic and environmental conditions, grouped by ZIP code, and consist of daily measurements from 2020 through 2025. The socioeconomic variables were population density, population, the number of 311 calls on a given day, and the number of drinking water fountains. The environmental variables were the maximum daily temperature, the volume of gas used for power, and the number of trees. The number of daily 311 calls and maximum daily temperature are special because they are temporal, meaning that their values change day to day. This introduces a novel element to our study. Based on the research we have conducted thus far, this is one of the first projects that integrates modeled temporal data when developing heat vulnerability indices. Data was collected from the following sources and organized using R.

Variable	Format	Temporal Extent
311 Calls recent	Individual calls logged	2010-2019, 2020-present (daily)
Natural gas consumption,	Consumption logged by ZIP code	2010 (not daily)
Steam Consumption	Consumption logged by ZIP code	2010 (not daily)
Street trees	Individual trees logged	2015
Drinking fountains	Program implemented in response to HVI values	2020
Population density	National density	2020 census
Daily temperature	Blanket value from JFK	2000-present

Model Specifics

Before we could begin training the models on the data, we had to create a risk signal that informed our models how to calculate the HVI score. The formula used for this risk signal intentionally excluded the official HVI scores created by the NYCDOHMH, because we did not want to bias our data with the numbers we would use to ground-truth it. For our risk signal calculation, we chose to use percentile-based binning to start, but are open to changing this later on because the resulting HVI score distributions are quite uniform, a feature discussed later on in this blog post. Each ZIP code's composite risk score, derived from our seven input variables, was ranked across all ZIP codes, and then divided into five equal-sized groups by percentile, with the lowest 20% assigned a score of 1 and the highest 20% assigned a score of 5.

Our three models were trained using an 80-10-10 test-train-validation split. Data was split using stratified random sampling, where the strata were HVI scores. The reasoning behind the split comes from maximizing the amount of data for training. Given that our dataset spans six years of daily ZIP-code-level observations, resulting in over 60,000 rows, the 10% validation and test sets still provide sufficiently large holdout samples (roughly 12,000 observations each) to evaluate model performance reliably. Following training and testing, we achieved the following validation results:

Decision Tree - Validation

	precision	recall	f1-score	support
1	0.91	0.84	0.88	2452
2	0.65	0.84	0.74	2451
3	0.60	0.57	0.58	2452
4	0.66	0.59	0.62	2452
5	0.87	0.83	0.85	2452
accuracy			0.73	12259
macro avg	0.74	0.73	0.73	12259
weighted avg	0.74	0.73	0.73	12259

Random Forest - Validation

	precision	recall	f1-score	support
1	1.00	1.00	1.00	2452
2	0.99	0.99	0.99	2451
3	0.99	0.99	0.99	2452
4	0.99	1.00	0.99	2452
5	1.00	0.99	1.00	2452
accuracy			0.99	12259
macro avg	0.99	0.99	0.99	12259
weighted avg	0.99	0.99	0.99	12259

XGBoost - Validation

	precision	recall	f1-score	support
1	1.00	1.00	1.00	2452
2	0.99	0.99	0.99	2451
3	0.99	0.99	0.99	2452
4	0.99	0.99	0.99	2452
5	1.00	0.99	0.99	2452
accuracy			0.99	12259
macro avg	0.99	0.99	0.99	12259
weighted avg	0.99	0.99	0.99	12259

Decision Tree - Test

	precision	recall	f1-score	support
1	0.92	0.84	0.88	2452
2	0.66	0.83	0.74	2452
3	0.60	0.58	0.59	2452
4	0.66	0.60	0.63	2452
5	0.88	0.83	0.85	2452
accuracy			0.74	12260
macro avg	0.74	0.74	0.74	12260
weighted avg	0.74	0.74	0.74	12260

Random Forest - Test

	precision	recall	f1-score	support
1	1.00	1.00	1.00	2452
2	0.99	0.99	0.99	2452
3	0.99	1.00	0.99	2452
4	0.99	0.99	0.99	2452
5	1.00	0.99	1.00	2452
accuracy			0.99	12260
macro avg	0.99	0.99	0.99	12260
weighted avg	0.99	0.99	0.99	12260

XGBoost - Test

	precision	recall	f1-score	support
1	1.00	1.00	1.00	2452
2	0.99	0.99	0.99	2452
3	0.99	0.99	0.99	2452
4	0.99	0.99	0.99	2452
5	0.99	1.00	0.99	2452
accuracy			0.99	12260
macro avg	0.99	0.99	0.99	12260
weighted avg	0.99	0.99	0.99	12260

Table 2: *Classification performance by model on validation and test sets.*

We also computed the pairwise disagreement between models to see how their predictions differed, and obtained the following results:

```
=== Model Agreement Analysis ===
Mean absolute difference (DT vs RF): 0.233
Mean absolute difference (DT vs XGB): 0.233
Mean absolute difference (RF vs XGB): 0.002
```

Table 3: *Pairwise disagreement between the three models.*

In sum, these results indicate that both the Random Forest and XGBoost models are good fits for our data and are very similar, while the decision tree model is very different and can be discarded. Because of its ability to handle repeat data values, we will continue with the XGBoost algorithm. This means that the following features will weigh most heavily when predicting HVI scores:

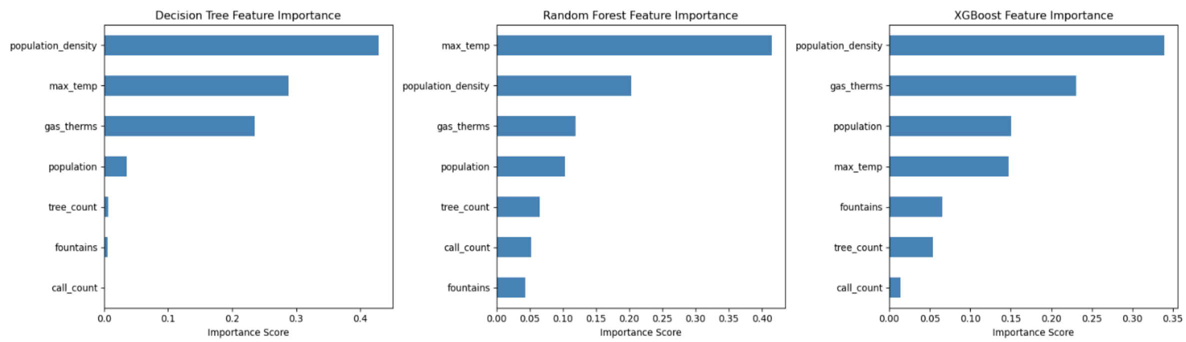

Figure 1: *Feature importance by model. Population density and gas usage are highly visible in the Decision Tree and XGBoost models, while maximum temperature leads in the Random Forest.*

Preliminary Results

Results for Hypothesis 1

Based on the data: yes, the three models trained on combined daily behavioral and environmental signals were able to predict established high-scoring HVI ZIP codes for 2024. Confusion matrices for each of the models and the ground-truth data clarify that the models, specifically the random forest and XGBoost ones, are especially good at correctly predicting higher HVI scores, whereas accuracy is lower at lower HVI scores. The decision tree model was very accurate and precise when predicting HVI scores of 4, but had lower predictive power for the other HVI scores. In line with the earlier results indicating high similarity between the random forest and XGBoost model, they have the same predictions for every single mean HVI score:

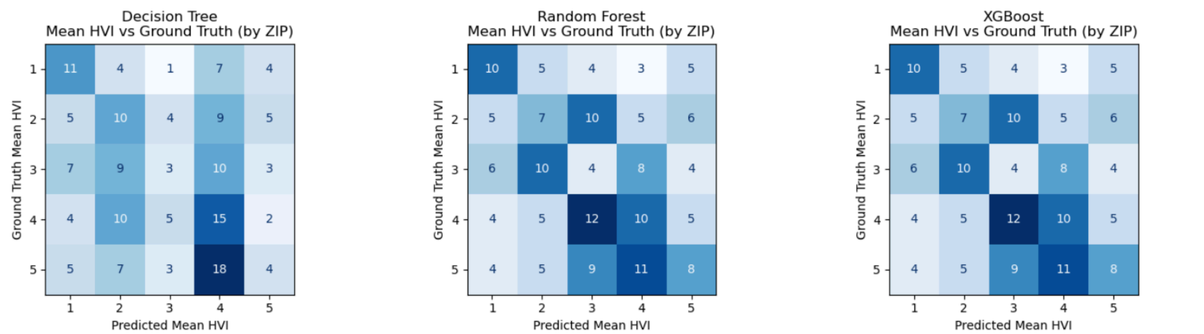

Figure 2: *Confusion matrices for each model as compared with the ground-truth. Random Forest and XGBoost usually make small mistakes, while the Decision Tree makes more scattered and less reliable errors. All models perform best at higher HVI levels (4–5), which is especially important because these are the neighborhoods where accurate identification matters most for resource allocation.*

The incorrect predictions are reflected in the distributions of HVI scores of the three models as compared to the ground-truth data:

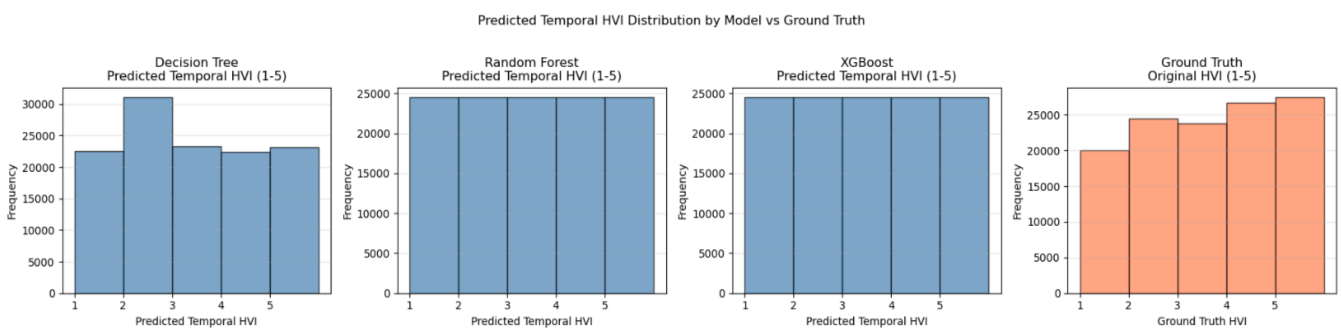

Figure 3: Predicted temporal HVI score distribution by model vs. ground-truth. *Random Forest and XGBoost models predict a more even spread of HVI levels, while the actual data is more concentrated in higher-risk categories.*

The random forest and XGBoost models predict a uniform distribution of HVI scores, while the decision tree does not. However, the ground-truth data show that frequency increases with HVI score. While the option to artificially induce a specific HVI score distribution does exist, we are hesitant to pursue this because we want to remain true to the variables that we have included. However, it is true that government resources are limited and that certain things must be prioritized over others. So, this model may need to enforce a less uniform distribution on the predictions for it to be more useful to policymakers.

Results for Hypothesis 2

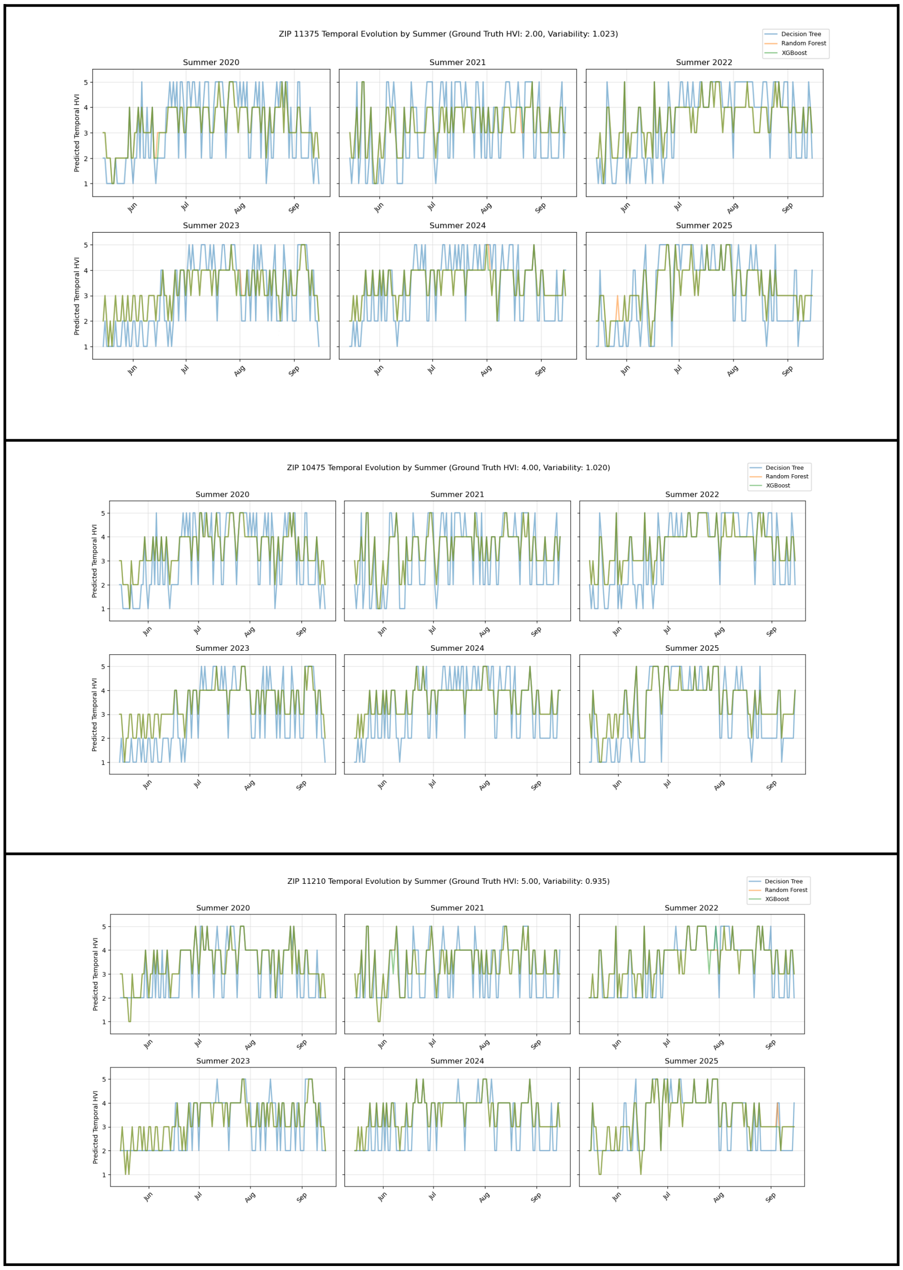

Figure 4: Temporal Variation in Heat Vulnerability Across 3 Most Variable ZIP Codes. *Heat vulnerability follows a clear seasonal arc across all three ZIP codes, rising through July and August before declining in September.*

Figure 4 shows a simple pattern: lower values are seen in June, rising into July and August, and slowly declining during September. This pattern suggests that the models are capturing meaningful temporal variations in heat vulnerability. The Decision Tree model shows abrupt fluctuations, whereas the XGBoost and Random Forest produce more consistent variations. It should also be noted that these predictions remain centered around each ZIP code's HVI baseline level, meaning that the models are capturing both temporal signals and neighborhood-differences in heat vulnerability. Indeed, the top 3 ZIP codes that yield these vulnerabilities are the Bronx, Queens, and Flatbush in Brooklyn.

Next Steps

Incorporate Data from 2018-2019

Our prototype was created using 5 years of data: from 2020-2025. Because our preliminary results show that HVI score does in fact vary with time, it could be informative to incorporate data from two additional years, especially to establish what HVI scores and distributions were like prior to the socioeconomic and environmental disparities exacerbated by the COVID-19 pandemic. Indeed, having this information could help our models to account for more long-term trends in HVI scores.

Add More Variables

One interesting outcome of our feature importance analysis was that the weights given to each feature varied widely between the models. Some of this is of course an artifact of the models' different architectures, but it could also be a symptom of underlying patterns in the data that could be illuminated if more data were included. Because NYCOpenData and U.S. Census records are very rich repositories, we could incorporate such datasets as median household income, racial demographics, mean English proficiency level, and percent participation in

Department of Sanitation programs. With our model established and verified as well-fitting, exploring the types of data we can include will make it more robust.

Conclusion

Our prototype model, trained with seven different datasets, suggests that we can reject our null hypothesis that models trained with temporal data will not be able to predict HVI scores in NYC ZIP codes. Additionally, our prototype model allows us to reject our null hypothesis that the temporal HVI score predictions will be the same as the mean HVI score predictions, indicating that HVI scores do in fact change over time. These preliminary findings suggest that integrating temporal machine learning approaches into existing HVI frameworks could improve real-time resource allocation and public health responses during extreme heat events.

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